# Strategy Report

Project: Precious Metal Premium Mean-Reversion Strategy

Last Update Time: 2026-04-22 08:30:50

Update Status: success

Update Message: □□□□□□□□

Best Strategy Summary

- Asset: silver
- Signal: BUY
- Holding Days: 20
- Final Capital: 41702.5
- Total Return %: 317.02
- Trades: 18

Core Findings

- BUY strategies outperform SELL strategies in the last 2 years.
- Silver BUY strategies show the strongest returns.
- Silver BUY 20D is currently the best-performing strategy.

# Backtest Summary Table

| asset | holding_days | signal | avg_return_pct | win_rate_pct | trades | max_return_pct | min_return_pct |
|---|---|---|---|---|---|---|---|
| gold | 5 | BUY | 0.82 | 62.5 | 16 | 8.46 | -8.59 |
| gold | 5 | SELL | 1.03 | 66.67 | 3 | 2.73 | -1.31 |
| gold | 10 | BUY | 1.18 | 62.5 | 16 | 13.09 | -7.42 |
| gold | 10 | SELL | -0.81 | 33.33 | 3 | 0.81 | -3.13 |
| gold | 20 | BUY | 3.69 | 68.75 | 16 | 22.58 | -2.97 |
| gold | 20 | SELL | -2.41 | 66.67 | 3 | 2.06 | -10.44 |
| silver | 5 | BUY | 2.62 | 55.56 | 18 | 15.28 | -3.68 |
| silver | 5 | SELL | -0.95 | 44.44 | 9 | 4.72 | -6.15 |
| silver | 10 | BUY | 4.48 | 72.22 | 18 | 22.93 | -4.51 |
| silver | 10 | SELL | -2.77 | 33.33 | 9 | 7.23 | -8.71 |
| silver | 20 | BUY | 9.1 | 83.33 | 18 | 61.92 | -9.57 |
| silver | 20 | SELL | -5.56 | 33.33 | 9 | 5.55 | -23.76 |

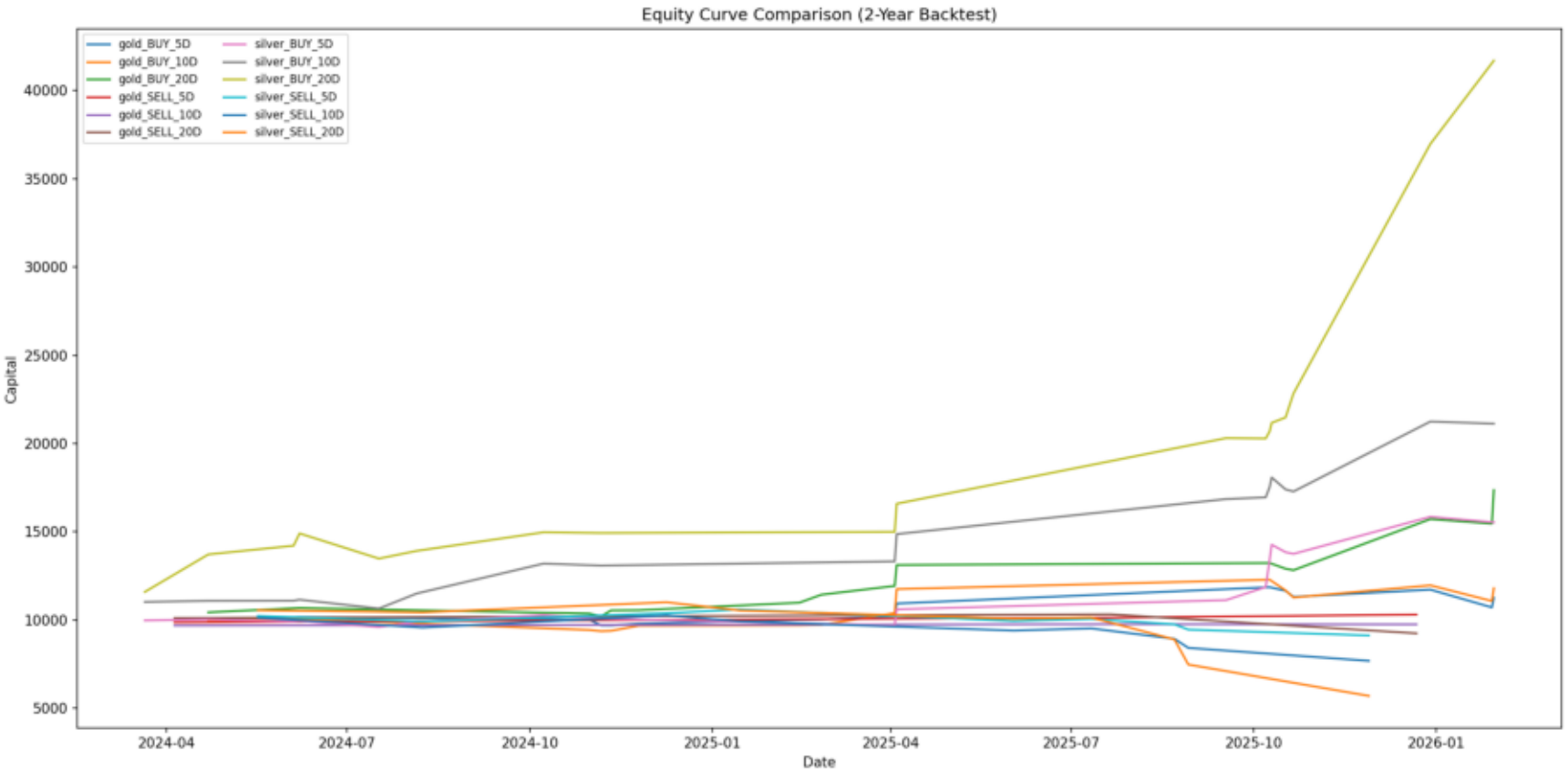
# Risk Metrics Table

| asset | signal | holding_days | trades | final_capital | total_return_pct | win_rate_pct | max_drawdown_p | volatility_pct | sharpe_ratio |
|---|---|---|---|---|---|---|---|---|---|
| silver | BUY | 20 | 18 | 41702.5 | 317.02 | 83.33 | -9.57 | 15.26 | 0.5967 |
| silver | BUY | 10 | 18 | 21129.45 | 111.29 | 72.22 | -4.51 | 7.38 | 0.607 |
| gold | BUY | 20 | 16 | 17364.68 | 73.65 | 68.75 | -4.46 | 6.59 | 0.5609 |
| silver | BUY | 5 | 18 | 15535.94 | 55.36 | 55.56 | -5.63 | 5.67 | 0.4625 |
| gold | BUY | 10 | 16 | 11802.7 | 18.03 | 62.5 | -9.84 | 5.4 | 0.2175 |
| gold | BUY | 5 | 16 | 11255.04 | 12.55 | 62.5 | -9.66 | 4.14 | 0.1984 |
| gold | SELL | 5 | 3 | 10308.0 | 3.08 | 66.67 | 0.0 | 2.1 | 0.4919 |
| gold | SELL | 10 | 3 | 9755.39 | -2.45 | 33.33 | -0.11 | 2.06 | -0.3923 |
| gold | SELL | 20 | 3 | 9245.33 | -7.55 | 66.67 | -10.44 | 6.97 | -0.3458 |
| silver | SELL | 5 | 9 | 9126.87 | -8.73 | 44.44 | -13.89 | 3.59 | -0.2652 |
| silver | SELL | 10 | 9 | 7687.32 | -23.13 | 33.33 | -25.13 | 5.04 | -0.5491 |
| silver | SELL | 20 | 9 | 5701.32 | -42.99 | 33.33 | -48.21 | 9.94 | -0.5595 |

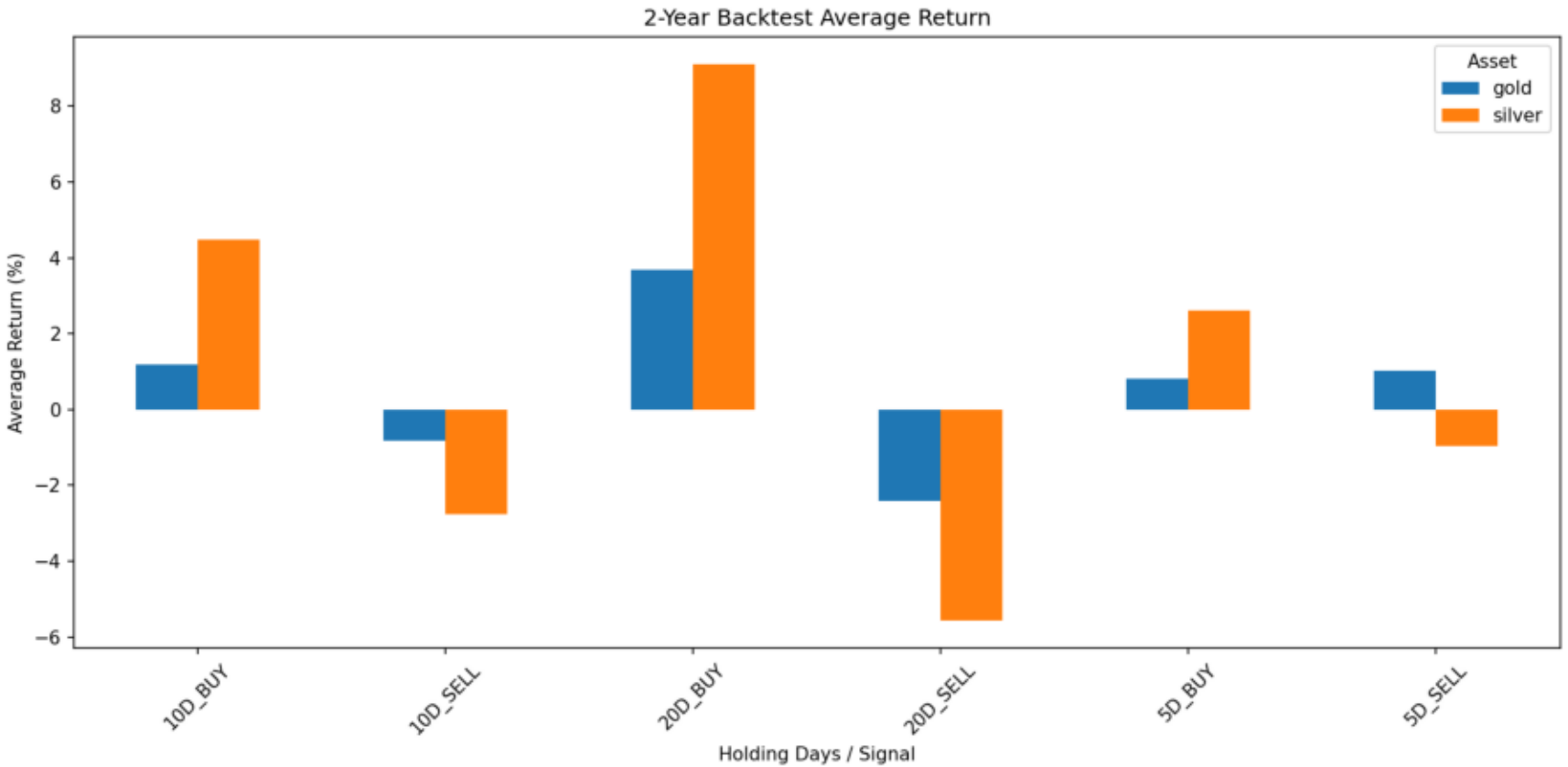
# Strategy Ranking Table

| asset | signal | holding_days | final_capital | trades | total_return_pct |
|---|---|---|---|---|---|
| silver | BUY | 20 | 41702.5 | 18 | 317.02 |
| silver | BUY | 10 | 21129.45 | 18 | 111.29 |
| gold | BUY | 20 | 17364.68 | 16 | 73.65 |
| silver | BUY | 5 | 15535.94 | 18 | 55.36 |
| gold | BUY | 10 | 11802.7 | 16 | 18.03 |
| gold | BUY | 5 | 11255.04 | 16 | 12.55 |
| gold | SELL | 5 | 10308.0 | 3 | 3.08 |
| gold | SELL | 10 | 9755.39 | 3 | -2.45 |
| gold | SELL | 20 | 9245.33 | 3 | -7.55 |
| silver | SELL | 5 | 9126.87 | 9 | -8.73 |
| silver | SELL | 10 | 7687.32 | 9 | -23.13 |
| silver | SELL | 20 | 5701.32 | 9 | -42.99 |

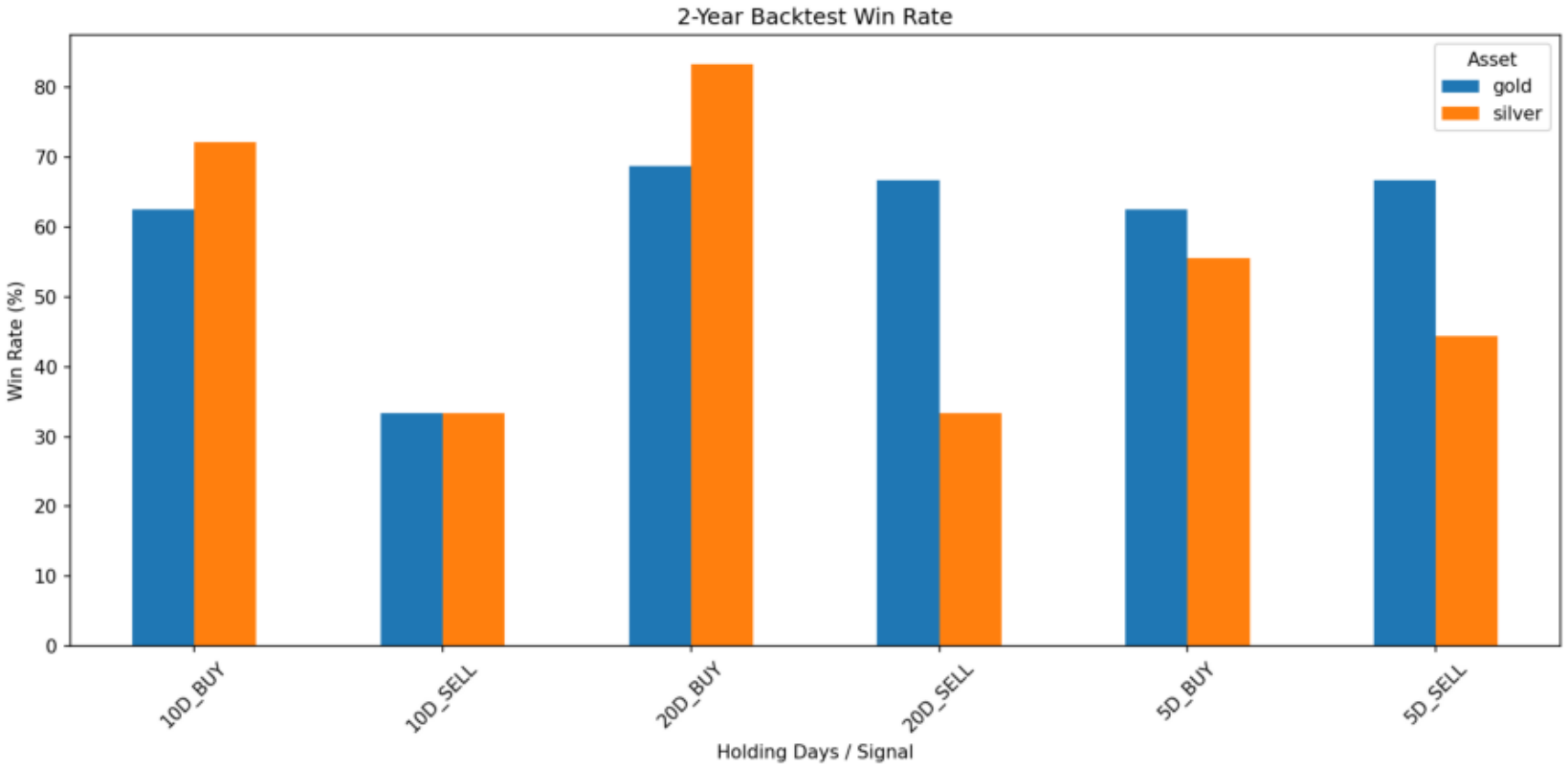
# Equity Curve Comparison



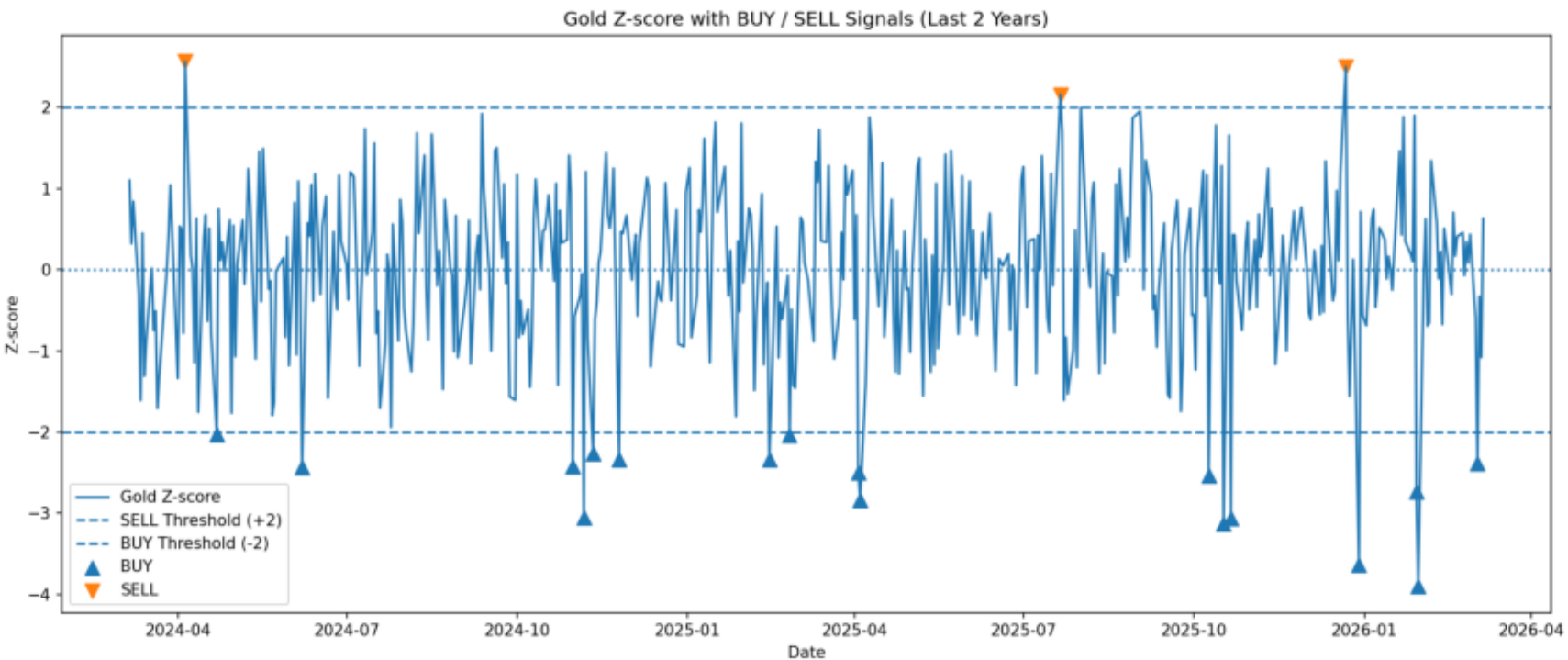
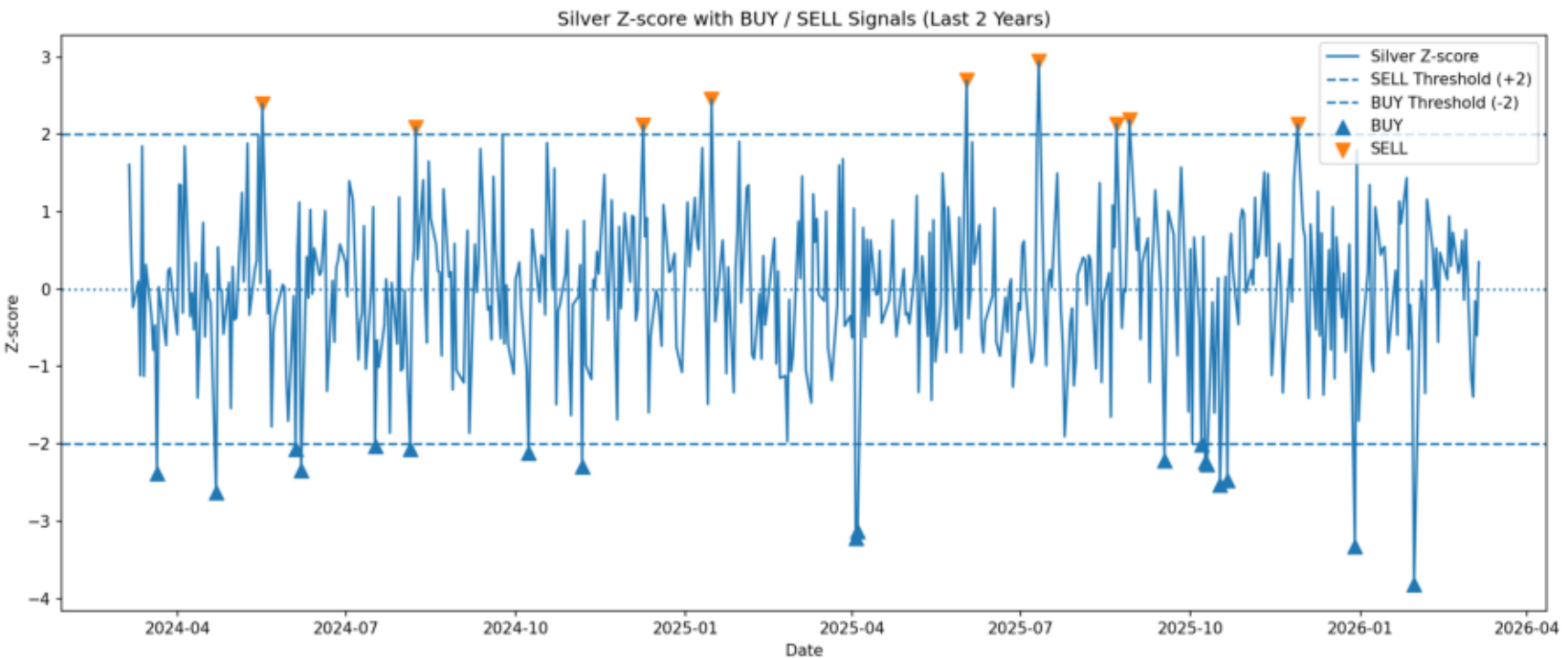
# Backtest Average Return



# Backtest Win Rate



# Gold Z-score Signals



# Silver Z-score Signals